Reproducing Spectral Shrunk Clustering: Faithful Stationary Points and Unreproduced Gains

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Abstract

This course report reimplements Spectral Shrunk Clustering (SSC) of Chang, Nie, Ma, Yang, and Zhou from the printed equations, without author code. After a typesetting correction, the faithful linear system is the stationary point of Equation (8), $A = S + 2\gamma\widetilde{L}$ (variant E8). Printed Equation (13) omits that factor of two (variant E13). Four of eight named datasets are usable; AR, BinAlpha, and MSRA50 were not obtained; UMIST archives contain 1012 or 575 images, not 564. Recovered AT&T 92×112 and CroppedYale 2414 files are extra loader versions; default ORL and YaleB loaders are unchanged. Shrinkage does *not* help: clustering G matches clustering F. Faithful E8 ORL accuracy is 53.3 ± 3.2 ($n = 50$) versus the published 76.0 ± 1.6; that 76% gain is *not* reproduced. YaleB uses $n = 10$ and USPS $n = 3$, not 50. A k-means++ reconstruction (B) is numerically closer on ORL/YaleB but is not the paper protocol. Row-normalized embeddings (C) are diagnostic only. A mutual kNN extension is negative or neutral. SPLS remains unimplemented. No new theorem is claimed.

1 Introduction

Spectral clustering [Shi and Malik, 2000, von Luxburg, 2007] builds a similarity graph, takes Laplacian eigenvectors, and runs K-means in that embedding. COMP3314 contrasts that neighbourhood geometry with PCA, which keeps high-variance axes, and with LDA, which uses class scatter. Chang et al. argue that a graph built on the original points x_i need not be the right regularizer for the embedded points f_i. Spectral Shrunk Clustering (SSC) therefore computes a normalized-cut embedding F, rebuilds a k-nearest-neighbour (kNN) graph on F, and solves a convex shrinkage problem for G before the final K-means [Chang et al., 2015].

This report keeps four tasks separate. First, we reimplement SSC from the equations and record every unspecified choice as an assumption. Second, we obtain datasets only from files with a URL, date, and SHA-256; missing names stay missing. Third, published ACC/NMI are labelled "Chang et al. (2015)" and are never copied into an Our Reproduction cell. Fourth, we replace the shrinkage graph by mutual kNN as a Category C extension, not as a new method, and report a negative or neutral difference. This is an HKU COMP3314 project, not a NeurIPS submission, and it states no novel theorem.

*HKU COMP3314 course report in the NeurIPS 2026 `preprint` style. It is *not* a NeurIPS submission. Group number is inferred from repository `COMP-3314-G23`. Member names and emails require confirmation and are not invented.

2 Background and SSC

Lloyd K-means is the last discretization in SSC and in the spectral baseline [Lloyd, 1982]. PCA+K-means keeps variance; unsupervised LDA+K-means of Ding and Li [2007] uses current cluster labels, never ground truth, and has rank at most $c-1$. Spectral clustering uses the symmetric normalized Laplacian of Equation (4) in Chang et al.,

$$L_n = I - D^{-1/2} A D^{-1/2}. \tag{1}$$

The paper sets $k = 5$ for every spectral method. We take δ^2 as the median of positive directed kNN squared distances (unspecified in the paper; Category B) and undirected edges as the union of the two directed relations [von Luxburg, 2007]. F holds the c eigenvectors for the smallest eigenvalues. Isolated vertices get $D_{ii}^{-1/2} = 0$. Eigenvectors are *not* row-normalized; Ng–Jordan–Weiss row-norm [Ng et al., 2001] is Category C even when ACC is higher. Local learning clustering [Wu and Schölkopf, 2006] and adaptive neighbours [Nie et al., 2014] are different methods. Pattern shrinking [Hou et al., 2013] is cited as SPLS (the source acronym is SLPS); we did not obtain Section 3 of that paper and keep SPLS as `NOT_IMPLEMENTED`.

Shrinkage. A second Gaussian kNN graph W is built on the rows of F. The printed convex problem is

$$\min_G \|G - F\|_{2,1} + \gamma \sum_{ij} W_{ij} \|g_i - g_j\|_2, \tag{2}$$

with $\|H\|_{2,1} = \sum_i \|h_i\|_2$ over rows. The sum is over ordered pairs. Equation (10) writes an embedding-dimension diagonal for S, which cannot sit in the $n \times n$ trace of Equation (9) or add to the $n \times n$ Laplacian. We use the sample-wise $S_{ii} = 1/(2\max(\|h_i\|_2, \varepsilon))$. Algorithm 1 line 10 replaces $\widetilde{L}$ by $\widetilde{W}$ and F by X; we do not execute that line.

Equation (8) versus Equation (13). For symmetric W, $\sum_{ij} \widetilde{W}_{ij} \|g_i - g_j\|_2^2 = 2\,\mathrm{Tr}(G^\top \widetilde{L} G)$. Differentiating (2) yields the stationary system $(S + 2\gamma\widetilde{L})G = SF$. Printed Equation (13) uses $\gamma\widetilde{L}$ instead of $2\gamma\widetilde{L}$, so it is the stationary point of (2) with graph coefficient $\gamma/2$. We treat that factor of two as a typesetting correction, not a contribution. Variant **E8** ($A = S + 2\gamma\widetilde{L}$) is the faithful reconstruction of printed Equation (8). Variant **E13** ($A = S + \gamma\widetilde{L}$) is the printed update and is Category B relative to Equation (8). A six-point CVXPY/CLARABEL check is in Appendix A. Default $G_0 = F$; the first all-zero residual uses a median pairwise floor rather than ε, which would freeze E13 at F. We stop on a relative Frobenius step of 10^{-6} or after 300 iterations; a small step is not a small fixed-point residual. Labels come from K-means on G and are never written back into W or F.

3 Reproduction setup

Results are tagged A (paper-faithful after the corrections above), B (unspecified reconstruction), or C (non-faithful diagnostic). A closer number does not promote C to A.

Default `load_dataset("ORL")` remains scikit-learn Olivetti 64×64 bilinearly resized to 32×32 [Samaria and Harter, 1994, Pedregosa et al., 2011], not native 92×112. Version `att92` is an extra AT&T archive (400 PGM, 92×112). JAFFE is the verified Zenodo TIFF set, 213 images, resized to 26×26 [Lyons et al., 1998, 2025]; redistribution is forbidden. Default YaleB is Deng Cai's `YaleB_32x32.mat` (2414×38); version `cropped2414` is a Wayback CroppedYale zip with 2414 non-ambient $(192, 168)$ images, not claimed pixel-identical to the mat [Georghiades et al., 2001, Lee et al., 2005]. USPS is the LIBSVM 9298×256 pair [Hull, 1994]. AR, BinAlpha, and MSRA50 were not obtained. UMIST/Sheffield files are 1012 uncropped or 575 cropped images, not the paper's 564; the default UMIST loader still raises `DatasetUnavailable` [Graham and Allinson, 1998]. No substitute collections were renamed.

Protocol R1 (Track B spirit): $k = 5$, $\gamma = 1$ frozen without labels, union kNN, median bandwidth, raw eigenvectors, $G_0 = F$, IRLS budget 300. F and G are solved once; K-means uses distinct seeds. We report mean$\pm$sample standard deviation over completed trials and never pad to 50. The paper's "50 times" / "best objective" / printed mean$\pm$std conjunction is left unresolved. ACC

Table 1: Hungarian ACC and geometric NMI (%). Paper cells are Chang et al. (2015), reported only. E8 is faithful A. k-means++ is B. Row-norm is C and is never the paper method. Run records: `results/repair_round1/`.

Set (n)	Metric	Paper (reported)	E8 (A)	k-means++ (B)	Row-norm (C)
ORL (50)	ACC	76.0±1.6	53.3±3.2	62.1±1.6	60.2±2.9
	NMI	88.6±1.5	76.2±1.8	80.3±0.7	80.1±1.2
JAFFE (50)	ACC	81.2±1.6	68.4±11.1	95.3±0.0	85.0±9.4
	NMI	84.3±1.6	83.0±5.7	94.1±0.0	91.8±4.1
YaleB (10)	ACC	49.8±1.6	19.5±1.5	25.2±0.6	26.7±1.8
	NMI	46.8±1.3	29.3±1.2	30.9±0.8	37.4±0.7
USPS (3)	ACC	75.5±1.9	N/A	63.8±1.7†	N/A
	NMI	79.8±1.6	N/A	77.5±4.0†	N/A

Mean±sample std over completed K-means trials; n is the actual count, not padded to 50. Paper cells are *not* Our Reproduction. E8 is the faithful stationary point of printed Eq. (8) ($A = S + 2\gamma\widetilde{L}$), union kNN, k=5, γ=1, raw eigenvectors, G_0=F, random K-means. k-means++ is a B reconstruction (paper unspecified; historical protocol is random). Row-norm is C (Ng–Jordan–Weiss), diagnostic only, never called the paper method. †USPS B cell is E13 (printed Eq. 13), not k-means++; E8 and row-norm were not run. AR, UMIST (no 564-image file), BinAlpha, MSRA50, and SPLS are N/A — not reproduced.

is Hungarian matching [Munkres, 1957]; NMI is the Strehl–Ghosh geometric mean [Strehl and Ghosh, 2002]. Hardware: 8 CPU cores, 16 GB RAM, Python 3.13.3, no GPU. Code: `https://github.com/SelinaMai-SYM/COMP-3314-G23.git` branch `v1`.

4 Results and analysis

Table 1 is the only main-text comparison. Paper columns quote Chang et al.; they are not Our Reproduction. Trial counts are actual: ORL/JAFFE $n = 50$, YaleB $n = 10$, USPS $n = 3$. We do not claim a 50-run YaleB or USPS protocol.

Shrinkage did not help. On the same embeddings, K-means(F) versus E13 K-means(G) is 53.9 ± 3.0 vs 53.7 ± 2.8 (ORL, $n = 50$), 68.5 ± 11.2 vs 68.6 ± 11.3 (JAFFE, $n = 50$), and 20.1 ± 1.2 vs 19.7 ± 1.1 (YaleB, $n = 10$). E8 is not better: 53.3, 68.4, 19.5. The convex post-process does not explain a large accuracy gain over spectral clustering on these files.

The ORL 76% result was not reproduced. Faithful E8 is 53.3 ± 3.2, 22.7 points below 76.0 ± 1.6. AT&T 92×112 bilinear E13 is 53.9 ± 3.8 ($n = 5$); nearest-neighbour resize is 58.8 ± 5.3 ($n = 5$). Neither is a 50-run recovery of 76%. Fixed bandwidth $\delta^2 = 1$ *hurts* (42.4 ± 1.8, $n = 50$). IRLS at $\gamma = 1$ on ORL does not meet the 10^{-6} step tolerance in 300 iterations.

Faithful versus numerically closest. The most faithful reconstruction is E8 (A). The numerically closest A/B reconstruction is *not* the same object: ORL k-means++ 62.1 ± 1.6 (B, still -13.9 points); JAFFE E13 68.6 ± 11.3 (B; k-means++ 95.3 ± 0.0 overshoots the paper and is not a reproduction of 81.2); YaleB k-means++ 25.2 ± 0.6 (B, $n = 10$, still -24.6); USPS E13 63.8 ± 1.7 (B, $n = 3$). Row-norm C is 60.2 / 85.0 / 26.7 and must not be reported as SSC. CroppedYale bilinear E13 is 18.0 ± 0.7 ($n = 10$), not an improvement over Deng Cai. Remaining gaps versus the paper, using faithful E8, are about -23 (ORL), -13 (JAFFE), and -30 (YaleB, $n = 10$) accuracy points.

5 Extension and discussion

The extension replaces union kNN W on F by mutual (AND) kNN with the same Gaussian weights frozen before IRLS [Brito et al., 1997, Maier et al., 2009]. Convexity of (2) is unchanged; no new theorem is claimed. At $\gamma = 1$, $k = 5$, mutual vs union ACC is 54.0 ± 2.8 vs 53.7 ± 2.8 (ORL, $n = 50$), 68.7 ± 11.3 vs 68.6 ± 11.3 (JAFFE, $n = 50$), and 20.0 ± 1.3 vs 19.7 ± 1.1 (YaleB, $n = 10$). Mutual graphs have fewer edges and more components. The outcome is negative or neutral, not a win, and it does not close the ORL or YaleB gap. USPS was not extended.

The modelling claim—regularize the embedding with a graph on the embedding—is independent of the typesetting errors, and it matches the COMP3314 distinction between pixel neighbourhoods and Laplacian coordinates. The published optimization story is weaker: Equation (10), Algorithm 1 line 10, and the Eq. (8)/Eq. (13) coefficient all change the linear system. Label-informed "best

results" on a γ grid is not a fair unsupervised tune; this round froze $\gamma = 1$. Four dataset names remain unmatched at the paper counts. SPLS, described as the second-best baseline, has no Our Reproduction numbers. K-means++ (B) moves ACC, which shows that the unspecified discretization matters, not that 2015 used k-means++.

Limitations in force: no author code; median bandwidth and $G_0 = F$ are assumptions; YaleB $n = 10$ and USPS $n = 3$; IRLS often unconverged on ORL/YaleB; Olivetti and Deng Cai defaults are not claimed sample-identical to 2015; face images collected by others are gitignored and not redistributed. Author-contact questions (coefficient, δ, G_0, K-means init, 564-image UMIST, BinAlpha seed, MSRA50 source, SPLS updates) are listed in the repository and were not emailed from this round.

6 Conclusion

We reimplemented SSC with an explicit E8/E13 split, recovered extra ORL and YaleB archives without changing default loaders, and compared published tables to completed runs only. Shrinkage does not help on the obtained files. Faithful E8 does not reproduce the ORL 76% accuracy. The most faithful reconstruction and the numerically closest reconstruction are not the same. The mutual-kNN extension is negative or neutral. SPLS is not implemented. The honest product is the protocol, the corrections, the missing data, and those negative findings.

Acknowledgments and Disclosure of Funding

HKU COMP3314 course report. No external funding is claimed. Names and emails require confirmation. Code is intended for `https://github.com/SelinaMai-SYM/COMP-3314-G23.git` on branch `v1`.

References

Jianbo Shi and Jitendra Malik. Normalized cuts and image segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 22(8):888–905, 2000. doi: 10.1109/34.868688.

Ulrike von Luxburg. A tutorial on spectral clustering. *Statistics and Computing*, 17(4):395–416, 2007. doi: 10.1007/s11222-007-9033-z.

Xiaojun Chang, Feiping Nie, Zhigang Ma, Yi Yang, and Xiaofang Zhou. A convex formulation for spectral shrunk clustering. In *Proceedings of the Twenty-Ninth AAAI Conference on Artificial Intelligence*, pages 2532–2538. AAAI Press, 2015. doi: 10.1609/aaai.v29i1.9606.

Stuart P. Lloyd. Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 28 (2):129–137, 1982. doi: 10.1109/TIT.1982.1056489.

Chris Ding and Tao Li. Adaptive dimension reduction using discriminant analysis and K-means clustering. In *Proceedings of the 24th International Conference on Machine Learning*, pages 521–528, 2007. doi: 10.1145/1273496.1273562.

Andrew Y. Ng, Michael I. Jordan, and Yair Weiss. On spectral clustering: Analysis and an algorithm. In *Advances in Neural Information Processing Systems*, volume 14, pages 849–856, 2001.

Mingrui Wu and Bernhard Schölkopf. A local learning approach for clustering. In *Advances in Neural Information Processing Systems*, volume 19, pages 1529–1536, 2006.

Feiping Nie, Xiaoqian Wang, and Heng Huang. Clustering and projected clustering with adaptive neighbors. In *Proceedings of the 20th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 977–986, 2014. doi: 10.1145/2623330.2623726.

Chenping Hou, Feiping Nie, Yuanyuan Jiao, Changshui Zhang, and Yi Wu. Learning a subspace for clustering via pattern shrinking. *Information Processing and Management*, 49(4):871–883, 2013. doi: 10.1016/j.ipm.2013.01.004.

F. S. Samaria and A. C. Harter. Parameterisation of a stochastic model for human face identification. In *Proceedings of the 1994 IEEE Workshop on Applications of Computer Vision*, pages 138–142, 1994. doi: 10.1109/ACV.1994.341300.

Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

Michael Lyons, Shigeru Akamatsu, Miyuki Kamachi, and Jiro Gyoba. Coding facial expressions with Gabor wavelets. In *Proceedings Third IEEE International Conference on Automatic Face and Gesture Recognition*, pages 200–205, 1998. doi: 10.1109/AFGR.1998.670949.

Michael J. Lyons, Miyuki Kamachi, and Jiro Gyoba. The Japanese female facial expression (JAFFE) dataset. Zenodo, 2025. Record 14974867; non-commercial research use, redistribution forbidden.

Athinodoros S. Georghiades, Peter N. Belhumeur, and David J. Kriegman. From few to many: Illumination cone models for face recognition under variable lighting and pose. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 23(6):643–660, 2001. doi: 10.1109/34.927464.

Kuang-Chih Lee, Jeffrey Ho, and David J. Kriegman. Acquiring linear subspaces for face recognition under variable lighting. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 27(5): 684–698, 2005. doi: 10.1109/TPAMI.2005.92.

Jonathan J. Hull. A database for handwritten text recognition research. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 16(5):550–554, 1994. doi: 10.1109/34.291440.

Daniel B. Graham and Nigel M. Allinson. Characterising virtual eigensignatures for general purpose face recognition. In Harry Wechsler, P. Jonathon Phillips, Vicki Bruce, Françoise Fogelman Soulié, and Thomas S. Huang, editors, *Face Recognition: From Theory to Applications*, NATO ASI Series F, pages 446–456. Springer, 1998. doi: 10.1007/978-3-642-72201-1_25.

James Munkres. Algorithms for the assignment and transportation problems. *Journal of the Society for Industrial and Applied Mathematics*, 5(1):32–38, 1957. doi: 10.1137/0105003.

Alexander Strehl and Joydeep Ghosh. Cluster ensembles — a knowledge reuse framework for combining multiple partitions. *Journal of Machine Learning Research*, 3:583–617, 2002.

M. R. Brito, E. L. Chávez, A. J. Quiroz, and J. E. Yukich. Connectivity of the mutual k-nearest-neighbor graph in clustering and outlier detection. *Statistics and Probability Letters*, 35(1):33–42, 1997. doi: 10.1016/S0167-7152(96)00213-1.

Markus Maier, Matthias Hein, and Ulrike von Luxburg. Optimal construction of k-nearest-neighbor graphs for identifying noisy clusters. *Theoretical Computer Science*, 410(19):1749–1764, 2009. doi: 10.1016/j.tcs.2009.01.009.

A Equation audit

This appendix records the locked E8/E13 split. It is a correction of Chang et al., not a claimed theorem.

Laplacian identity. For symmetric nonnegative W and combinatorial $\widetilde{L} = \widetilde{D} - \widetilde{W}$,

$$\sum_{ij} \widetilde{W}_{ij} \|g_i - g_j\|_2^2 = 2\,\mathrm{Tr}(G^\top \widetilde{L} G).$$

A two-point check with $G = (0.2, 0.8)^\top$, $W_{01} = W_{10} = 1$, and distance 0.6 gives $\sum_{ij} W_{ij}\|g_i - g_j\|_2^2 = 0.72$, $\sum_{ij} \widetilde{W}_{ij}\|g_i - g_j\|_2^2 = 0.6$, and $\mathrm{Tr}(G^\top \widetilde{L} G) = 0.3$.

Gradient of Equation (2)**.** For $h_k \neq 0$ and unused self-loops,

$$\frac{\partial}{\partial g_k}\|h_k\|_2 = 2S_{kk}(g_k - f_k), \qquad \frac{\partial}{\partial g_k}\sum_{ij} W_{ij}\|g_i - g_j\|_2 = 4(\widetilde{L}G)_k.$$

Setting the gradient to zero and dividing by 2 yields $(S + 2\gamma\widetilde{L})G = SF$. Printed Equation (13) omits the 2. The same factor appears in the IRLS majorizer of Equation (15) in the paper, because that quadratic uses $\sum_{ij} \widetilde{W}_{ij}\|g_i - g_j\|_2^2 = 2\,\mathrm{Tr}(G^\top \widetilde{L} G)$.

Independent convex check. On a synthetic chain with $n = 6$, $c = 2$, $\gamma = 0.8$, seed 1, CVXPY 1.9.3 / CLARABEL 0.11.1 reports status `optimal`. E8 matches the minimizer of printed Equation (2) (objective 4.021064576828 vs 4.021064576768; largest absolute entry difference 1.07×10^{-6}). E13 matches the $\gamma/2$ instance (objective 3.843869772479 vs 3.843869772520) and does *not* minimize printed Equation (2) (value 4.536864561036 at that G). A two-block graph repeats the split. These are solver-audit values, not clustering scores, and they are not copied from Chang et al.

Four unequal scalars. The unsmoothed objective, the ε-smoothed objective, the IRLS quadratic surrogate, and the recomputed fixed-point residual are four different numbers. A small IRLS step can coexist with a large residual.

B Dataset files

Checksums were recorded on 2026-10-20 and live in the named manifests. Raw images are gitignored. Default loaders are unchanged.

ORL. Default: figshare Olivetti (`benchmarks/orl/manifest.yaml`). Extra `att92`: Cambridge AT&T 400 images at 92×112 (`manifest_att92.yaml`).

JAFFE. Zenodo record 14974867 (`benchmarks/jaffe/manifest.yaml`). Non-commercial research only; no redistribution.

YaleB. Default: Deng Cai `YaleB_32x32.mat` (`manifest.yaml`). Extra `cropped2414`: Wayback CroppedYale zip, 2414 non-ambient $(192, 168)$ PGM (`manifest_cropped.yaml`). Not claimed pixel-identical to the mat. Live UCSD zips returned HTTP 404.

USPS. LIBSVM `usps.bz2` / `usps.t.bz2` (`benchmarks/usps/manifest.yaml`).

Unavailable at paper counts. AR: Ohio State HTTP 503, Purdue 404; no 840-image subset was built. UMIST archives on disk are 1012 uncropped or 575 cropped images, not 564 (`manifest_wayback.yaml`, `manifest_figshare_cropped.yaml`). BinAlpha: no 26×30 file; Binary Alphadigits was not substituted. MSRA50: no 1799×12 file; MSRA-B/10K/TD500 were not renamed.

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   Justification: Sections 2–4 and Appendix A specify the solver variants, protocol R1, metrics, manifests, and assumptions. Table 1 reports mean$\pm$sample standard deviation with the actual trial count n. Code, configs, and `runs/` records are in the course repository `https://github.com/SelinaMai-SYM/COMP-3314-G23.git` on branch `v1`. We do not claim that a third party can recover unpublished author binaries.

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